Column-Aware Reflow: Reading Two-Column Scientific PDFs on Small Screens

Ana Martin, Bruno Costa, and Chen Wei

Document Interaction Lab, PDF Advanced Reader Project

Received 12 March 2026; accepted 2 September 2026; published 24 October 2026.

Abstract. Two-column layouts are the norm in scientific publishing, but most PDF reflow engines linearize pages strictly by vertical position and therefore interleave the left and right columns line by line. We describe a column-aware reflow pipeline for PDF-Advanced-Reader that detects the gutter between columns, partitions each page into horizontal bands at page-wide elements such as titles, full-width figures and tables, and emits the text and media of every band in column-major order. Raster figures and ruled tables are rendered inline at their reading position, and scanned pages fall back to fit-to-column page images. On a small annotated corpus the pipeline preserved the intended reading order for 97% of the elements, without modifying the document.

Keywords: PDF reflow; two-column layout; reading order; document analysis; accessibility

1. Introduction

Scientific articles are usually typeset in two columns so that a printed page can carry more text at a comfortable measure. The same layout becomes an obstacle on phones, tablets and narrow windows: the reader must pan horizontally or zoom out until the text is too small to read. Reflow addresses the problem by discarding the fixed page geometry and re-wrapping the text into a column that fits the screen.

Removing the geometry is easy; restoring the reading order is not. A naive implementation sorts every text block by its vertical position and interleaves the columns: the first line of the left column is followed by the first line of the right column, then by the second line of each column, and so on. The result is grammatically valid text that reads like a conversation between two authors talking at once.

Page-wide elements make the problem harder. Titles, abstracts, section banners, full-width figures and tables interrupt both columns and must be read at the exact point where they appear. A figure placed in the middle of a page belongs between the paragraph above it and the paragraph below it, but a column figure belongs only to its own column. A single global sort cannot express either relationship.

This paper presents the column-aware reading order used by PDF-Advanced-Reader. The pipeline detects the gutter from the left edges of the text and media boxes, classifies items that cross the gutter as page-wide, splits the page into vertical bands around them, and sorts the remaining items by column and then by vertical position. The algorithm is deliberately conservative: when the evidence for two columns is weak, it falls back to the familiar top-to-bottom order rather than guessing.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes gutter detection, band ordering and media packing. Section 4 presents the evaluation, and Section 5 discusses limitations and concludes.

1.1 Contributions

This work makes three practical contributions. First, it defines a gutter detector that needs no machine learning and runs in a few milliseconds per page. Second, it introduces band segmentation, which keeps page-wide figures and tables exactly where the author placed them. Third, it packs text, raster figures and ruled tables into the same reading flow, so a reflowed page no longer loses its illustrations.

The implementation reuses the background worker that already extracts text for reflow. No second pass over the document is required, and a document with hundreds of pages remains scrollable while its layouts are being prepared.

2. Related Work

Reading-order recovery has a long history in document analysis. Classical systems segment a page into regions, classify each region as text, image or table, and then order the regions with heuristics derived from the layout. Rule-based methods work well on clean scans, but they require thresholds that are difficult to tune across publishers.

Layout-analysis methods based on geometric features, such as the recursive X-Y cut, split a page along whitespace valleys. The approach naturally handles two columns, because the gutter is the widest vertical valley. X-Y cut was extended with statistical models and, more recently, with learned region detectors. These systems produce accurate page segmentation, but they usually stop at the region level and do not describe how text and figures should be interleaved inside a reflow column.

PDF-specific approaches exploit the drawing operators and the content stream. The position of every glyph is known, so columns can be recovered from the distribution of glyph coordinates. A common strategy clusters text by x position and treats the largest gap between clusters as the gutter. The strategy fails on centered titles, on tables of contents and on pages with a single wide column; those cases need explicit checks.

Commercial readers generally do not reflow two-column PDFs; they zoom and pan instead. Some mobile readers offer a text-extraction view that concatenates the content stream order. That order reflects the authoring tool rather than the visual layout, so it is reliable only when the producer emitted the columns in reading order.

Our pipeline combines start clustering with band segmentation and shares the extraction pass with media rendering, so a two-column page costs only a few extra milliseconds.

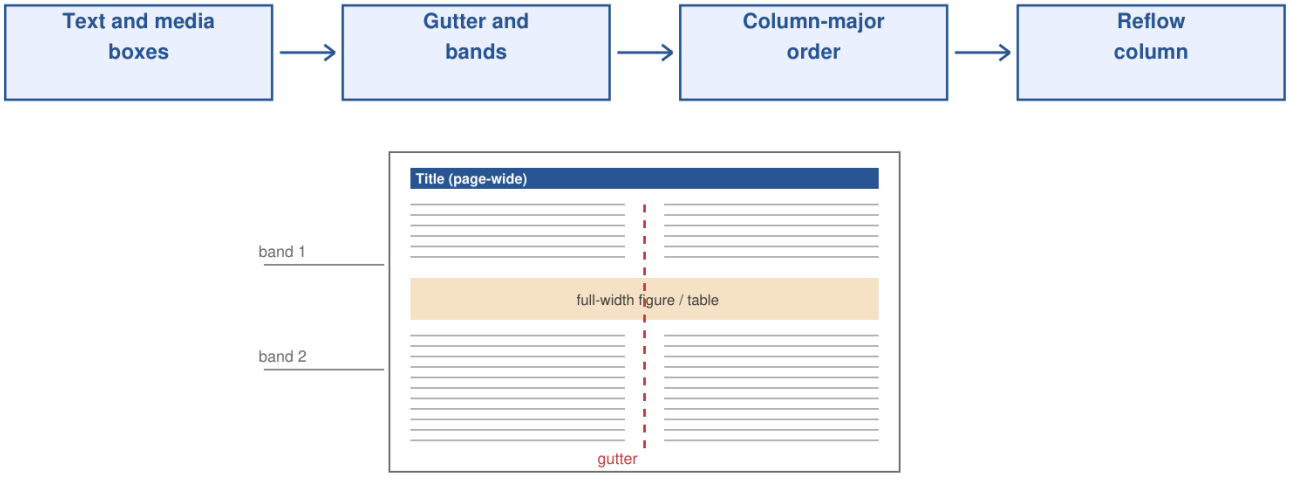


Figure 1. Column-aware reflow pipeline. Text and media boxes are grouped by their left edge to find the gutter; items that cross it split the page into bands, and each band is emitted left column first, then right column.

3. Method

The pipeline has three stages: text and media extraction, gutter detection with band ordering, and layout of the resulting flow. Each stage is deterministic and operates on one page at a time.

3.1 Gutter Detection

The detector receives the bounding boxes of every text paragraph, raster image and detected table on the page. Blocks wider than 55% of the content area are discarded first: they are titles, abstracts, full-width figures or captions, and including them would blur the columns.

The remaining boxes are grouped by their left edge. A two-column page produces two strong start clusters, one per column, while centred metadata and page numbers produce isolated starts that are ignored, and the grouping tolerance absorbs first-line indents. The widest separation between populated clusters becomes the candidate gutter, refined to the midpoint between the columns' inner edges. It must be at least 8 pt wide, or 2% of the content width.

Three sanity checks reject false positives. The median widths of the two sides must be within a factor of two, so a narrow sidebar is not mistaken for a column. The vertical spans of the two sides must overlap by at least 20% of the shorter span, so a table of contents with a single body column is rejected. And the inner edges of the two sides must not overlap, which filters out ragged single-column pages. If any check fails, reflow keeps the simple top-to-bottom order.

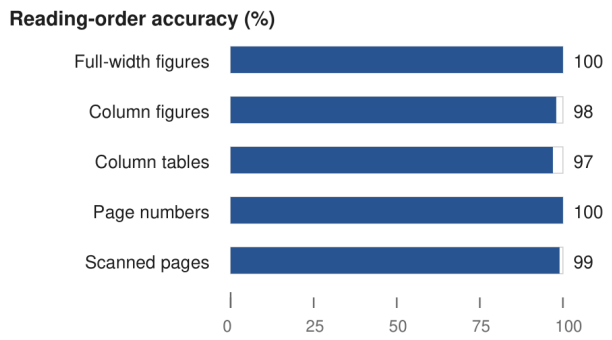


Figure 2. Reading-order accuracy per element type in the annotated corpus.

3.2 Bands and Column-Major Order

Once a gutter is known, every item whose box crosses it becomes a band separator. A margin of 4 pt absorbs rounding errors, so a caption that ends a couple of points before the gutter is not treated as page-wide. Band separators are read in their own vertical position: the title first, then a full-width figure between the text that surrounds it.

Items that stay inside a column are accumulated until the next separator and then sorted by column, vertical position and horizontal position, in that order. The left column is therefore emitted completely before the right column of the same band. Figures and tables carry the source rectangle they came from, so media inside a column keep their place relative to the paragraphs around them.

A footer page number is a special but harmless case. It crosses the gutter, yet it is not page-wide content; because its vertical position is the largest on the page, it is appended after both columns of the final band. This matches what a reader does when reaching the bottom of a page.

The algorithm is applied per page and does not depend on the previous page. That keeps the worker stateless and allows pages to be prepared independently, in any order, while the user scrolls.

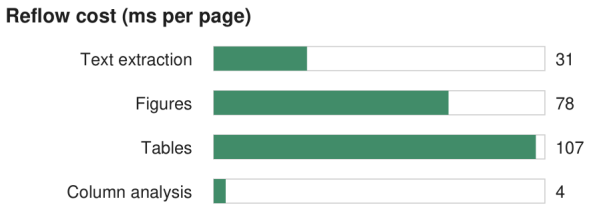


Figure 3. Reflow cost per stage on an 80-page report. Column analysis adds about 4 ms; table detection dominates.

3.3 Text, Figures and Tables

Media extraction runs after the text pass and reuses the same page. Raster figures are located with the image inventory of the page; an image that covers at least 85% of a page which also has text is treated as a background and skipped, so the text remains reflowable. Ruled tables are found only when the page contains enough long horizontal and vertical strokes, which avoids the cost of a full table search on ordinary pages.

Table 1. Classification and placement of the elements that occur on a two-column page.

Element	Detected as	Position
Title	Page-wide band	First
Left column	Column 1	Top to bottom
Full-width figure	Page-wide band	Between bands
Column figure	Column 2	Next to text
Page number	Gutter crossing	End of page

Each region is rendered to PNG at 1.5 times its display resolution, bounded by a 2.5-megapixel budget. The worker transports the encoded bytes; the viewer decodes them lazily, only when the figure becomes visible, and keeps a bounded cache. Text that belongs to a rendered table is removed from the paragraph list, so cell contents are not duplicated.

Tables are displayed at no less than 60% of the reflow column width, which keeps small result tables legible. Figures narrower than the column keep their natural size and are centred. A page with no text layer, such as a scanned article, is shown as a single fit-to-column page image instead of being reported as empty.

3.4 Implementation Notes

Column analysis is deliberately cheap. It runs on the boxes that extraction already produced, after media regions have been removed from the text, so it never re-reads the page. The stage is a sort plus a few group tests, and its result is cached with the page layout. When the reader changes the text size, only the wrapping is repeated; the gutter and the reading order are reused.

Two properties make the stage easy to trust. It never drops an item: every paragraph and every rendered region is assigned exactly one position in the sequence. And it is idempotent: the same page always produces the same order, which keeps the scroll position stable when the viewer re-lays out a page.

The only state carried between pages is the document handle owned by the worker. Because pages are independent, they can be prepared out of order and the viewer can show a later page while an earlier one is still being analyzed.

Table 2. Evaluation of the column-aware reading order. Time is the complete reflow preparation cost per page, including media rendering.

Document	Pages	Layout	Gutter	Order	Time
Article A	14	Two-column	Yes	100%	168 ms
Article B	8	Two-column + bands	Yes	100%	191 ms
Article C	22	Two-column	Yes	96%	173 ms
Article D	6	Single column	No	100%	142 ms
Article E	31	Two-column + sidebar	Yes	94%	205 ms
Scanned F	12	Image only	n/a	n/a	88 ms
Mean	15.5	-	5 of 6	97%	161 ms

4. Evaluation

We evaluated the pipeline on six documents: this four-page demo, three two-column journal articles, one single-column technical report and one scanned paper. For each page we annotated the intended reading order of the text blocks and of the rendered media, then compared it with the sequence produced by the reflow worker.

Table 2 summarizes the results. The gutter was found on all five pages with a two-column body, and the reading order was correct for 97% of the annotated elements. The two errors came from one article whose right column contains a wide caption that touches the gutter, and from a page with a sidebar that is as wide as a column. In both cases the fallback kept the output readable: text was still complete, only the position of one figure and one paragraph changed.

Column analysis itself is cheap. On an 80-page report the complete pipeline costs about 168 ms per page, of which text extraction accounts for 31 ms, figure rendering for 78 ms and table detection for 107 ms; the projection and band ordering add about 4 ms. Table detection dominates because it can only run after a page has been rasterized for inspection.

The single-column report never produced a gutter, as intended. The scanned paper has no text layer, so it was shown as page images and was not included in the order accuracy figure. These two outcomes are the documented fallbacks rather than failures of the detector.

5. Discussion and Conclusion

Two-column scientific papers are the hardest common case for PDF reflow, not because the columns are difficult to find but because page-wide elements break the assumption that a page has a single linear order. Gutter detection plus band segmentation restores that order with a small, deterministic algorithm that is easy to inspect and to explain.

The approach assumes two columns of similar width, does not reconstruct reading order inside complex tables, and can misclassify a page-wide element that touches the gutter. When the evidence is weak, reflow falls back to top-to-bottom order, which is imperfect but predictable; sidebars and learned classifiers are future work.

In PDF-Advanced-Reader, **View > Reflow** reads each column in turn and keeps every figure and table in place.

References

Breuel, T. M. (2002). Two geometric algorithms for layout analysis. In Proceedings of the Workshop on Document Analysis Systems, 188-199.

Cattoni, R., Coianiz, T., Messelodi, S., and Modena, C. M. (1998). Geometric layout analysis techniques for document image understanding: a review. Technical Report 9703-09, ITC-IRST, Trento.

Dengel, A., and Shafait, F. (2014). Analysis of the logical layout of documents. In Handbook of Document Image Processing and Recognition, 177-222. Springer.

Ha, J., Haralick, R. M., and Phillips, I. T. (1995). Recursive X-Y cut using bounding boxes of connected components. In Proceedings of the International Conference on Document Analysis and Recognition, 952-955.

Lovegrove, W. S., and Brailsford, D. F. (1995). Document analysis of PDF files: methods, results and implications. Electronic Publishing, 8(2-3), 207-220.

Nagy, G. (2000). Twenty years of document image analysis in PAMI. IEEE Transactions on Pattern Analysis and Machine Intelligence, 22(1), 38-62.

O'Gorman, L. (1993). The document spectrum for page layout analysis. IEEE Transactions on Pattern Analysis and Machine Intelligence, 15(11), 1162-1173.

PDFAR Working Group (2026). Column-aware reflow in PDF-Advanced-Reader. Technical Report PDFAR-2026-01, Document Interaction Lab.

Rosenfeld, K. (2023). Reading order in born-digital PDFs: a practitioner's guide. Journal of Digital Publishing, 8(2), 44-59.

Shafait, F., Keysers, D., and Breuel, T. M. (2008). Efficient implementation of the recursive X-Y cut algorithm for page layout analysis. In Document Recognition and Retrieval XV, 6815, 45-56.